

Google Cloud Engineering Certificate – Practical Lab Answers

PRACTICAL TEST 1: Create and Configure a Virtual Machine

Aim:

To create a Compute Engine virtual machine with appropriate configuration, firewall rules, and high availability settings.

Requirements:

- Google Cloud Project
- Compute Engine enabled
- Google Cloud Console
- Google Cloud CLI

Procedure:

1. Login to Google Cloud Console.
2. Navigate to Compute Engine → VM Instances.
3. Click Create Instance.
4. Configure machine type: e2-standard-4.
5. Select region us-central1 and zone us-central1-a.
6. Choose OS image Ubuntu 22.04 LTS.
7. Enable HTTP and HTTPS traffic.
8. Enable Automatic Restart.
9. Click Create.

Program (CLI):

```
gcloud compute instances create web-server-vm --machine-type=e2-standard-4 --image-family=ubuntu-2204-lts --image-project=ubuntu-os-cloud --zone=us-central1-a --tags=http-server,https-server --restart-on-failure
```

Output:

A VM instance is created and accessible through HTTP/HTTPS.

Result:

Virtual Machine successfully deployed with secure and reliable configuration.

PRACTICAL TEST 2: Instance Template and Managed Instance Group

Aim:

To deploy multiple VM instances using an instance template and configure autoscaling.

Requirements:

- Google Cloud Project
- Compute Engine enabled
- Google Cloud CLI

Procedure:

1. Create an Instance Template.
2. Configure VM specifications.
3. Create Managed Instance Group.
4. Enable autoscaling based on CPU utilization.
5. Configure load balancing.

Program (CLI):

```
gcloud compute instance-templates create web-template --machine-type=e2-medium --image-family=debian-11 --image-project=debian-cloud
```

```
gcloud compute instance-groups managed create web-group --template=web-template --size=2 --zone=us-central1-a
```

```
gcloud compute instance-groups managed set-autoscaling web-group --max-num-replicas=5 --target-cpu-utilization=0.6 --zone=us-central1-a
```

Output:

Multiple VM instances automatically scale based on CPU usage.

Result:

Highly scalable infrastructure created using Managed Instance Groups.

PRACTICAL TEST 3: Secure Cloud Storage Bucket

Aim:

To create a Cloud Storage bucket with restricted access and secure permissions.

Requirements:

- Google Cloud Project
- Storage Admin permissions
- Google Cloud CLI

Procedure:

1. Navigate to Cloud Storage → Buckets.
2. Click Create Bucket.
3. Select region us-central1.
4. Disable Public Access.
5. Assign IAM permissions to service account.

Program (CLI):

```
gsutil mb -p PROJECT_ID -l us-central1 gs://image-storage-bucket/
```

```
gcloud storage buckets update gs://image-storage-bucket --public-access-prevention=enforced
```

```
gsutil iam ch serviceAccount:web-app@PROJECT_ID.iam.gserviceaccount.com:objectAdmin gs://image-storage-bucket
```

Output:

Secure storage bucket created with restricted access.

Result:

Images and application files are securely stored in Cloud Storage.

PRACTICAL TEST 4: Cloud SQL MySQL Database Setup

Aim:

To create a Cloud SQL MySQL instance and configure secure database access.

Requirements:

- Google Cloud Project
- SQL Admin permissions
- Google Cloud CLI

Procedure:

1. Navigate to SQL → Create Instance.
2. Select MySQL database engine.
3. Configure instance version MySQL 8.0.
4. Select region us-central1.
5. Configure authorized networks.
6. Create database and user.

Program (CLI):

```
gcloud sql instances create mysql-instance --database-version=MYSQL_8_0 --  
tier=db-f1-micro --region=us-central1
```

```
gcloud sql databases create appdb --instance=mysql-instance
```

```
gcloud sql users create appuser --instance=mysql-instance --  
password=StrongPassword123
```

Output:

MySQL database instance created successfully.

Result:

Secure Cloud SQL database configured for application use.

PRACTICAL TEST 5: Secure Private VPC Network

Aim:

To create a custom VPC network with internal communication and restricted internet access.

Requirements:

- Google Cloud Project
- Network Admin permissions
- Google Cloud CLI

Procedure:

1. Create a custom VPC network.
2. Create subnet in us-central1.
3. Launch VM without external IP.
4. Configure firewall rules.
5. Enable Private Google Access.

Program (CLI):

```
gcloud compute networks create secure-vpc --subnet-mode=custom
```

```
gcloud compute networks subnets create secure-subnet --network=secure-vpc  
--range=10.0.0.0/24 --region=us-central1
```

```
gcloud compute instances create internal-vm --zone=us-central1-a --  
subnet=secure-subnet --no-address
```

```
gcloud compute firewall-rules create allow-internal --network=secure-vpc --  
allow=tcp,udp,icmp --source-ranges=10.0.0.0/24
```

Output:

VM instances communicate internally without internet exposure.

Result:

Secure internal network infrastructure successfully implemented.